Year 7
Mathematics

Unit 3: Equations, Functions and Formulae

3.1 Simplifying Algebraic Expressions

Simplifying Algebraic Expressions

STARTER · 5 MIN

Date: _____ **Title: Simplifying Algebraic Expressions**

LEARNING OBJECTIVES

- To be able to simplify expressions by collecting like terms

STARTER *Answer in your book. No calculators.*

1. Work out $3 + 5 - 2$
2. Work out 7×4
3. Work out $-3 + 8$
4. Work out $6 - 9$

Starter — Answers

STARTER

Tick or correct in green pen

1. **6**

2. **28**

3. **5**

4. **−3**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify like terms in an expression	☐
SILVER	I can collect like terms to simplify an expression	☐
GOLD	I can simplify expressions with powers and products of terms	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Simplifying Algebraic Expressions

INTRODUCTION · 3 MIN

Like terms have the same letters and powers. Only like terms can be collected.

Term

A number, a letter, or both multiplied.

$5x$ 3 $7ab$ y^2

Like terms

Same letter part.

$3x$ and $5x$ $2y^2$ and y^2

Unlike terms

Different letter part: keep apart.

$3x$ and $2y$ a and a^2

Expression

Terms joined by + or −.

$3a + 2b - 4$

Examples 1 and 2 — Collecting Like Terms

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Like terms: same letter Add the numbers in front

Example 1

Simplify

$3x + 5x$

Example 2

Simplify

$7a - 2a$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Add the numbers in front

$$3x + 5x = 8x$$

Tip: 3 lots of *x* plus 5 lots of *x*.

Example 2

Subtract the numbers in front

$$7a - 2a = 5a$$

Tip: the letter stays the same.

Task 1

PRACTICE · 9 MIN

BRONZE *Simplify*

Hint Like terms: same letter Add the numbers in front

1. $4x + 2x$

2. $9y - 3y$

3. $a + a + a$

4. $6b + b$

5. $8m - 5m + 2m$

★ $10p - 4p - p$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $6x$

2 $6y$

3 $3a$

4 $7b$

5 $5m$

★ $5p$

Examples 3 and 4 — Two Kinds of Term

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Group like terms first The sign goes with the term after it

Example 3

Simplify

$4b + 3 + 2b$

Example 4

Simplify

$3a + 2b + 4a - b$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Collect the *b* terms, then the numbers

$$4b + 2b + 3 \quad = 6b + 3$$

Tip: 6*b* and 3 are unlike: stop.

Example 4

Group the like terms

$$3a + 4a + 2b - b \quad = 7a + b$$

Tip: the sign belongs to the term after it.

Task 2

PRACTICE · 9 MIN

SILVER

Simplify by collecting like terms

Hint Group like terms first The sign goes with the term after it

1. $6y + 2 - 3y + 5$

2. $5x + 4 + 2x - 1$

3. $2a + 3b + 4a + b$

4. $7m + 2n - 3m - n$

5. $8 + 3p - 2 + p$

★ $4c - 2d + c + 5d$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $3y + 7$

2 $7x + 3$

3 $6a + 4b$

4 $4m + n$

5 $4p + 6$

★ $5c + 3d$

Examples 5 and 6 — Powers and Multiplying Terms

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint x^2 and x are unlike $b \times b = b^2$

Example 5

(2 marks)

Simplify

$5x^2 + 2x + 3x^2$

Example 6

(2 marks)

Simplify

(a) $3b \times 2b$

(b) $b \times b$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

x^2 and x are unlike terms

$$5x^2 + 3x^2 + 2x \quad = 8x^2 + 2x$$

Tip: $x^2 = x \times x$, not the same as x.

Example 6

Multiply numbers, then letters

(a) $3 \times 2 \times b \times b \quad = 6b^2$

(b) $b \times b \quad = b^2$

Tip: $b \times b = b^2$, not $2b$.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: simplify fully

Hint x^2 and x are unlike $b \times b = b^2$

1. $2a + 3b + 5 - a + 4b - 2$

2. $4y^2 + 3y - y^2 + y$

3. $4a \times 3$

4. $2x \times 5y$

5. $3m \times 4m$

★ Why is $3x + 2y$ not equal to $5xy$?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $a + 7b + 3$

2. $3y^2 + 4y$

3. $12a$

4. $10xy$

5. $12m^2$

★ **3*x* and 2*y* are unlike terms: they cannot be collected.**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $5x + 3x - x$

SILVER Simplify $4a + 7 + 2a - 3$

GOLD Why is $3x + 2y \neq 5xy$?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **$7x$**

SILVER **$6a + 4$**

GOLD **$3x$ and $2y$ are unlike terms, so they cannot be collected.**

Exam Practice — Collecting Like Terms

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint
Like terms: same letter
Add the numbers in front

Question 1

(2 marks)

Simplify

(a) $m + m + m + m$ (1)

.....

(b) $6p - 2p$ (1)

.....

Question 2

(2 marks)

Simplify

(a) $5x + 3x - 2x$ (1)

.....

(b) $4y + 3 + y$ (1)

.....

Exam Practice — Simplifying Expressions

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Group like terms first
The sign goes with the term after it

Question 3

(2 marks)

Simplify fully

$6a + 3b - 2a + 5b - 4$

.....

Question 4

(2 marks)

Simplify fully

$7x + 2y - 3x - 5y$

.....

Exam Practice — Products and Powers

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint x^2 and x are unlike $b \times b = b^2$

Question 5

(2 marks)

Simplify

(a) $3b \times 2b$ (1)

.....

(b) $b \times b$ (1)

.....

Question 6

(3 marks)

Simplify fully

$4x^2 + 3x - x^2 + 5 - x$

Sara writes $3x + 2y = 5xy$.

Explain why she is wrong.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $4m$ — B1

(b) $4p$ — B1

Tip: count the letters.

Question 2

(a) $6x$ — B1

(b) $5y + 3$ — B1

Tip: $5y$ and 3 are unlike terms.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$6a - 2a = 4a$, $3b + 5b = 8b$ — **M1**

$4a + 8b - 4$ — **A1**

Tip: keep the − 4 at the end.

Question 4

$7x - 3x = 4x$, $2y - 5y = -3y$ — **M1**

$4x - 3y$ — **A1**

Tip: 2 − 5 = −3.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $6b^2$ — B1

(b) b^2 — B1

Tip: $b \times b = b^2$.

Question 6

$3x^2 + 2x + 5$ — B2

B1 for two terms correct

Unlike terms cannot be added — B1

Tip: x^2 and x are unlike terms.